

# NATURAL LANGUAGE PROCESSING

## Foundations, Web Scraping, Sentiment Analysis & Applications

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2026-I



UNIVERSIDAD DISTRITAL  
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# Outline

- 1 Foundations and Basic Concepts
- 2 Web Scraping and Data Collection
- 3 Sentiment Analysis
- 4 Applications and Integration



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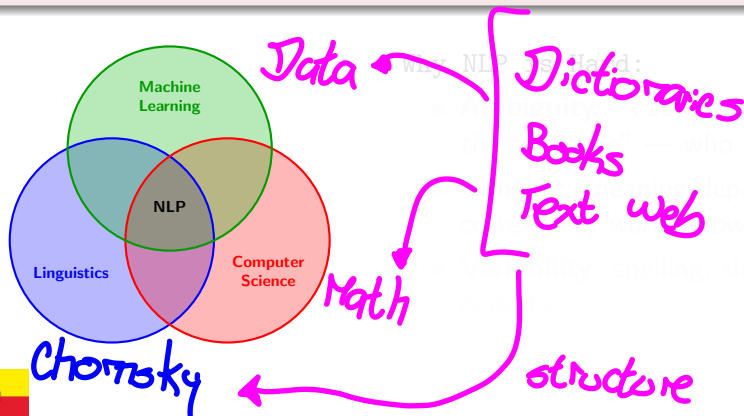


# What is Natural Language Processing?

*understand / interpretation like on human*

## Definition

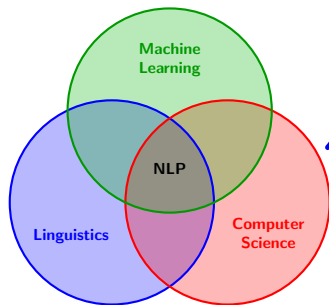
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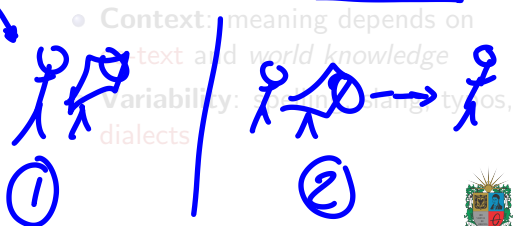
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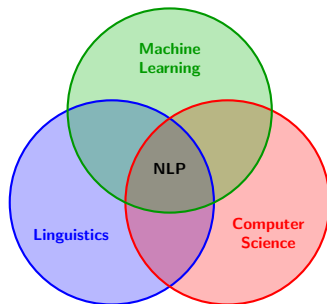
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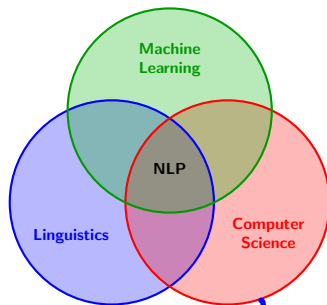
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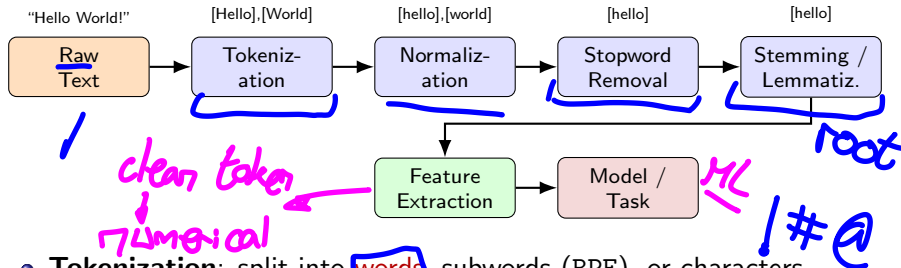
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*How a machine can do it?*



# The NLP Processing Pipeline



- **Tokenization:** split into words subwords (BPE), or characters
- **Normalization:** lowercase, accent removal, punctuation stripping
- **Stemming:** running → run; **Lemmatization:** better → good  
(grammatically correct root)

• **stopword:** the, is, at, on  
↳ repetition



## Bag-of-Words & TF-IDF

Bow

- 
- hello  
world  
LD
- king
- queen
- man
- woman
- analogies
- $w_1$   
 $w_2$   
 $w_3$
- $w=3$
- occurrences  
↓  
frequency
- Word2Vec 2-D projection



## Text Representation: From Words to Vectors

## Bag-of-Words &amp; TF-IDF

- Doc  $\rightarrow \mathbf{v} \in \mathbb{R}^{|V|}$ ; **order ignored**
- $\text{tfidf}(t, d) = \text{tf} \cdot \log \frac{|D|}{\text{df}_t}$
- Downweights **common**, upweights **rare** terms

1) el perro muerde al niño  
 log  $\rightarrow$  Frequency

## Word Embeddings

- Dense  $\mathbb{R}^d$ ; semantic similarity  $\approx$  distance
- king - man + woman  $\approx$  queen

2) helb  
 1) 4D  $\rightarrow$  3

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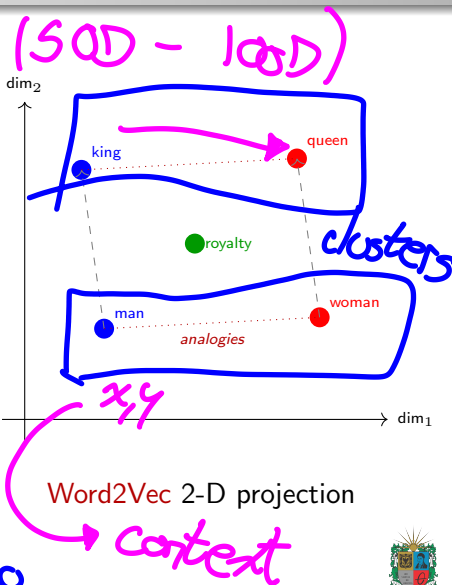
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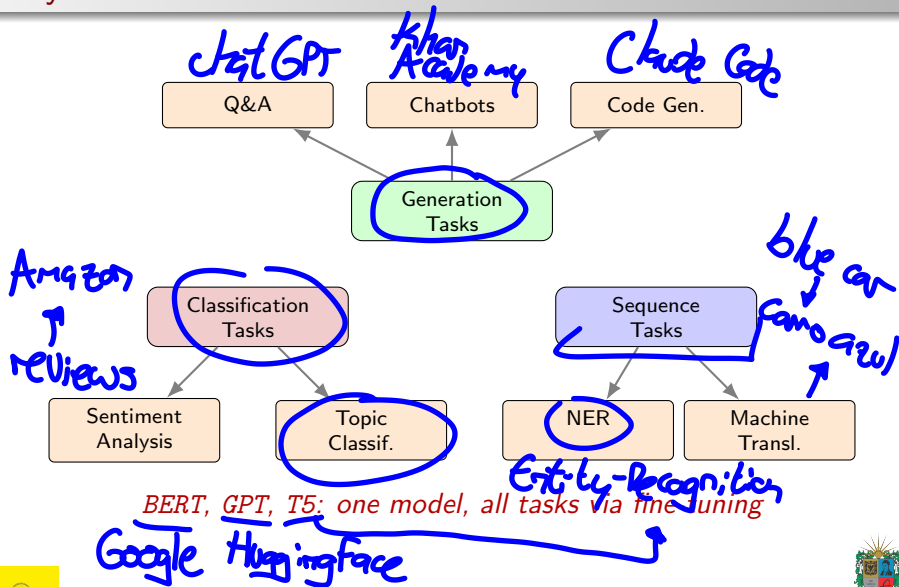
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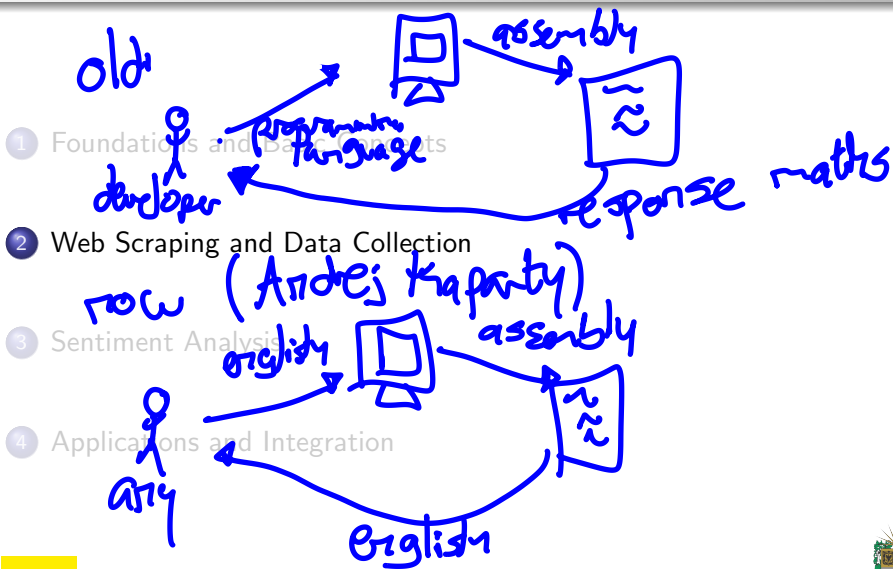
*caso*  $\leftrightarrow$  *edificio*  
*caso*  $\leftrightarrow$  *caso*



# Key NLP Tasks Overview



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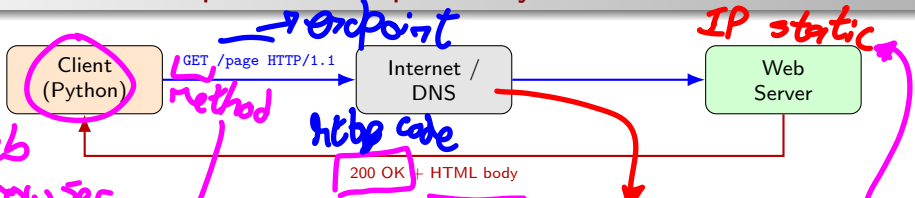


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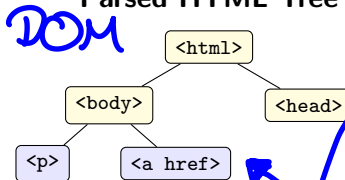
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# The HTTP Request — Response Cycle



## Parsed HTML Tree



**AJAX**

C → POST  
R → GET ✓  
P → PATCH  
D → DELETE

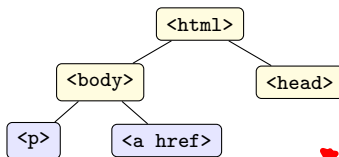
`<html/>`  
`<head> </head>`  
`<body>`  
`<p>...</p>`  
`<a href>...</a>`



# The HTTP Request — Response Cycle



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• Status: 200 OK | 403 Forbidden | 429 Rate Limited

• Dynamic page: JS renders after HTTP response

→ need Selenium

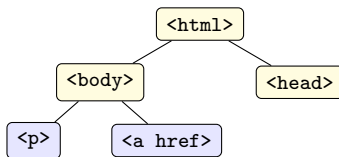
Free quota



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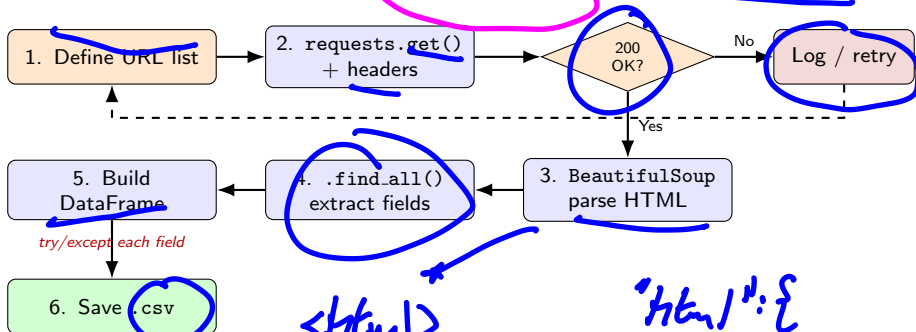
static web pages

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- **Dynamic pages:** JS renders *after* HTTP response

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# Scraping Workflow with requests + BeautifulSoup



- `time.sleep(1--5)` between requests (politeness delay)
- Always respect robots.txt terms of Service

Handwritten HTML tags:

```

<html>
<body>
<div>
<section>
<a>
</a>

```

Handwritten CSS selectors:

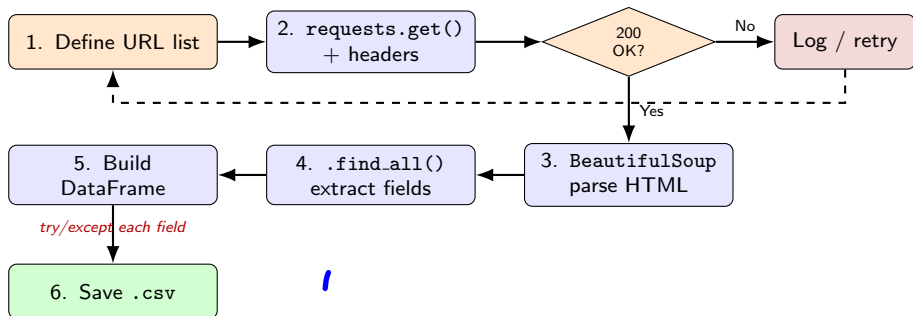
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```



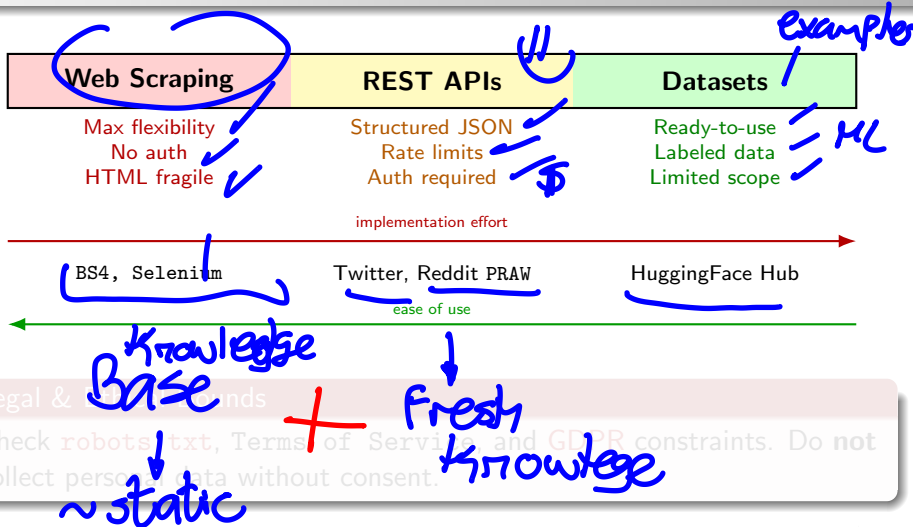
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# Scraping vs. APIs vs. Ready Datasets



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| Web Scraping                               | REST APIs                                       | Datasets                                      |
|--------------------------------------------|-------------------------------------------------|-----------------------------------------------|
| Max flexibility<br>No auth<br>HTML fragile | Structured JSON<br>Rate limits<br>Auth required | Ready-to-use<br>Labeled data<br>Limited scope |
| implementation effort                      |                                                 |                                               |
| BS4, Selenium                              | Twitter, Reddit PRAW                            | HuggingFace Hub                               |
| ease of use                                |                                                 |                                               |

## Legal & Ethical Bounds

Check robots.txt, Terms of Service, and GDPR constraints. Do **not** collect personal data without consent.

~contract

NYTimes



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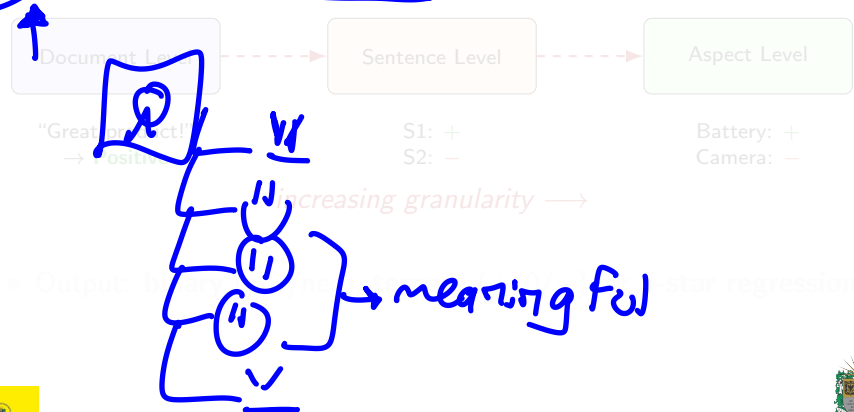
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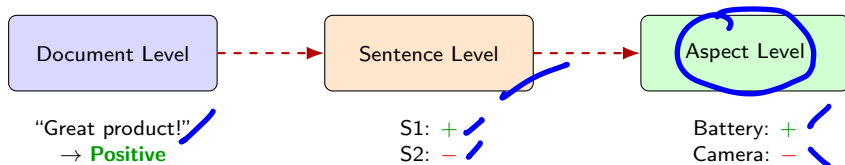
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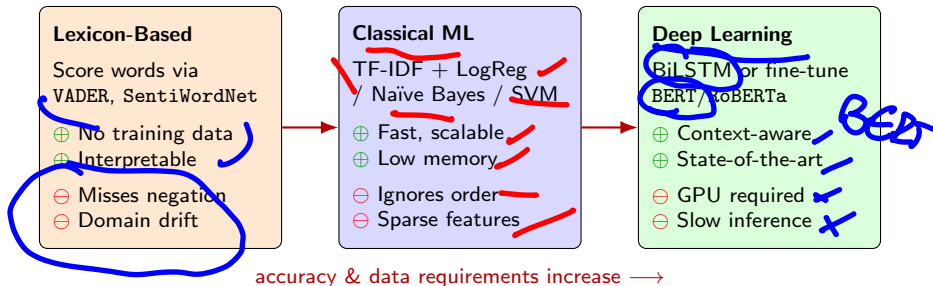


*increasing granularity →*

Output: **binary** (pos/neg), **ternary** (+/0/-), or **5-star regression**



# Three Approaches: Lexicon, ML, Deep Learning



AI → NLP



# Evaluation: *Metrics and Confusion Matrix*

## Beyond Accuracy

**Accuracy** is misleading on **imbalanced** data. Always report per-class metrics.

- Precision =  $\frac{TP}{TP + FP}$
- Recall =  $\frac{TP}{TP + FN}$
- F1 =  $\frac{2 \cdot P \cdot R}{P + R}$
- F1-macro: average over all classes

## Confusion Matrix

|           |              |            |            |
|-----------|--------------|------------|------------|
|           |              | high prec. |            |
|           |              | Pos (pred) | Neg (pred) |
| high rec. | Pos (actual) | TP         | FN         |
|           | Neg (actual) | FP         | TN         |



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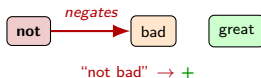
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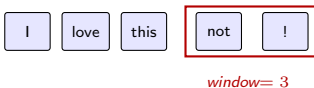
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- **Negation:** “*Not bad*” → **positive**
- **Sarcasm:** “*Great, another Monday*”
- **Domain shift:** medical  $\neq$  movie vocabulary
- **Multilingual:** cultural conventions differ
- **Subjectivity:** “*Film runs 120 min*” has no sentiment

## Negation Scope



## Context Window



## Practical Tip

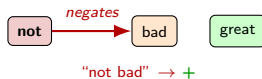
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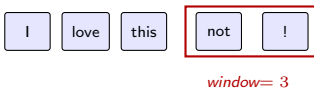
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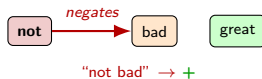
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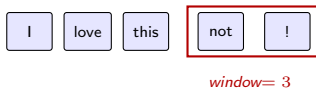
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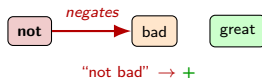
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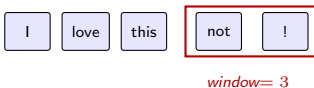
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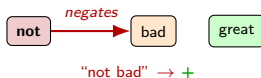
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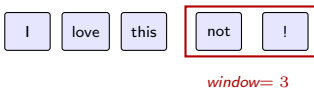
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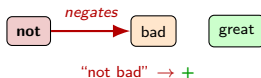
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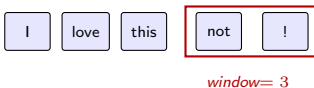
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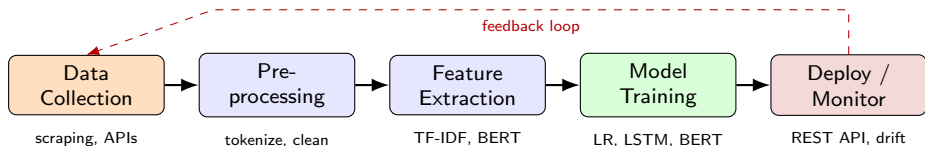


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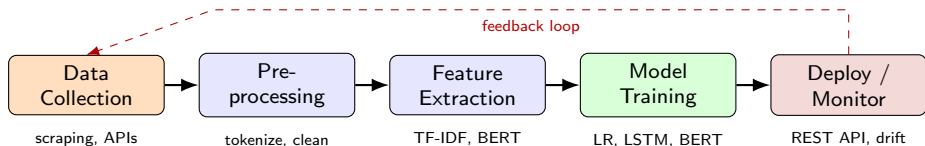


## Real-World Applications:

- Chatbots & virtual assistants
- Machine translation (DeepL)
- Search engine ranking
- NER in clinical records
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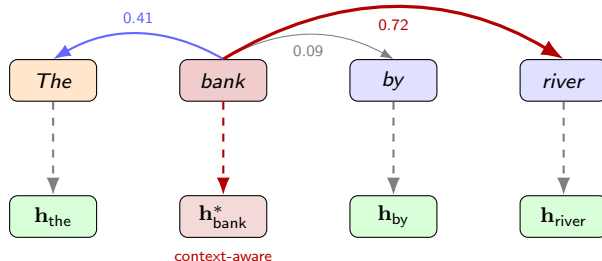


# Transformer Models: Self-Attention

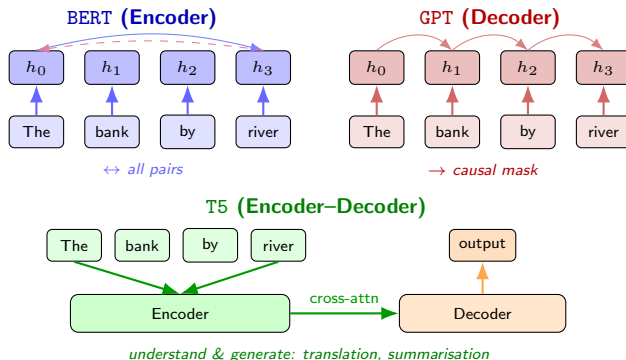
## Key Innovation (Vaswani *et al.*, 2017)

Replace recurrence with **attention**: every token simultaneously queries **all** other tokens, capturing **long-range context** in parallel.

*“bank” attends most to “river” → river bank, not financial bank*



# Transformer Families: Architecture Comparison



## BERT pre-training

MLM: predict masked tokens

"The [MASK] by the river"

## GPT pre-training

CLM: predict next token

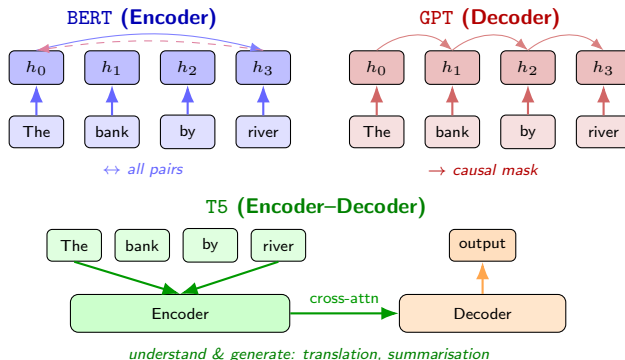
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## T5 pre-training

Span corruption: mask spans

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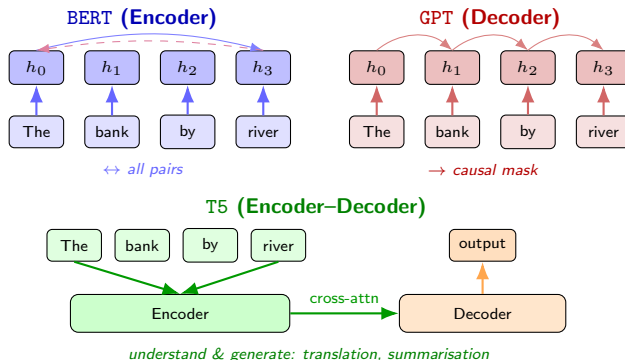
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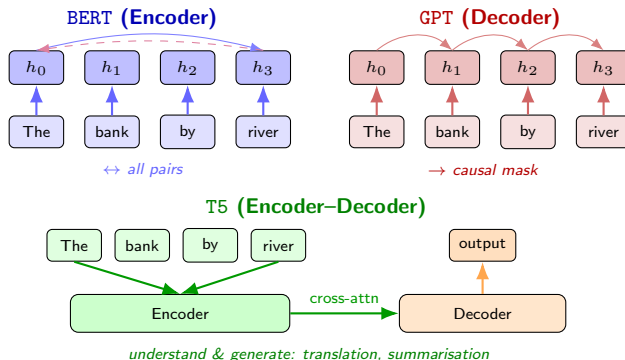
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# Sentence Embeddings & Semantic Search

## Encoding a Sentence

- Feed full sentence through BERT
- Pool [CLS] token or mean-pool hidden states
- Output:  $\mathbf{v} \in \mathbb{R}^{768}$  — a semantic fingerprint

## Cosine Similarity

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

1  $\approx$  same meaning

0  $\approx$  unrelated



Semantic search in embedding space

*This is the Embed + Vector DB step  
in RAG*



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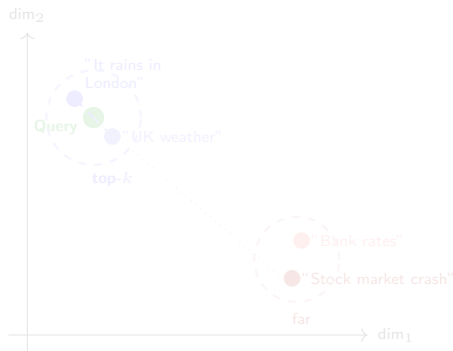
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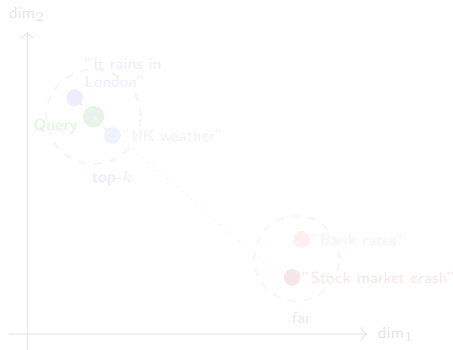
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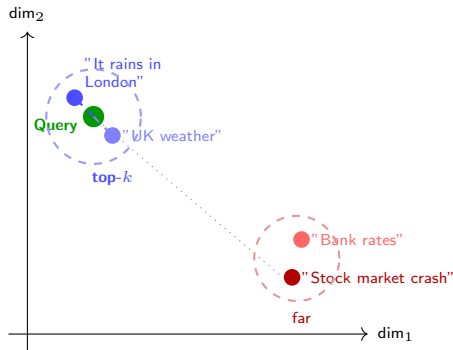
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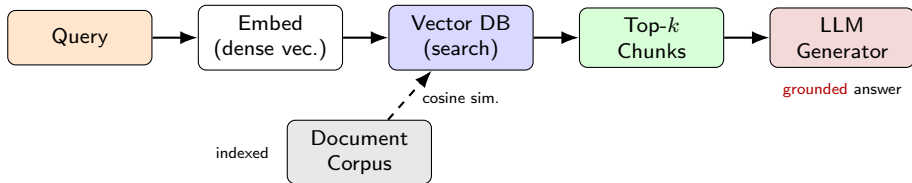
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in RAG*



# Retrieval-Augmented Generation (RAG)

## Motivation

LLMs hallucinate. **Grounding** responses in a **verified document store** dramatically reduces factual errors.



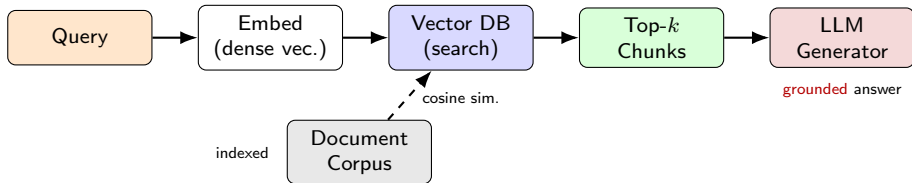
- Tools: LangChain, LlamaIndex, Chroma, FAISS
- Enables domain-specific assistants without full retraining



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# Current Challenges in NLP

- **Hallucination:** LLMs generate **plausible but false** content
- **Bias:** stereotypes inherited from **training corpora**
- **Low-resource languages:**  $\sim 7,000$  languages; most resources are **English-centric**
- **Interpretability:** SHAP, LIME, Attention maps as post-hoc explanations
- **Computational cost:** GPT-4  $\approx$  \$200M to train; LoRA/PEFT for efficient fine-tuning
- **Privacy:** text often contains PII — **GDPR** compliance required

## Workshop Connection

Build a complete pipeline: scrape *reviews*  $\rightarrow$  *preprocess*  $\rightarrow$  *train sentiment classifier*  $\rightarrow$  *deploy* as a Streamlit app with live **inference**.



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# Outline

- 1 Foundations and Basic Concepts
- 2 Web Scraping and Data Collection
- 3 Sentiment Analysis
- 4 Applications and Integration



# Thanks!

## Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming>

